

Canonical Form of LPP

- If the objective function is of the maximization type, then all constraints must be of the “less than or equal to” ($\leq$) type.
- If the objective function is of the minimization type, then all constraints must be of the “greater than or equal to” ($\geq$) type.
- All decision variables must be non-negative, i.e., $x_j \geq 0$.

In the canonical form, the LPP is expressed as:

Maximize: $c^T x$

Subject to: $Ax \leq b, x \geq 0$.

Or

Minimize: $c^T x$

Subject to: $Ax \geq b, x \geq 0$.

Standard Form of LPP

- All decision variables must be non-negative, i.e., $x_j \geq 0$.
- The right-hand side of each constraint equation must be non-negative, i.e., $b_i \geq 0$.
- All constraints are expressed as equalities (except for the non-negativity restrictions).
- The system must remain linear, i.e., all constraints and the objective function must involve linear expressions.

In the standard form, the LPP is expressed as:

Maximize / Minimize: $\mathbf{c}^T \mathbf{x}$

Subject to: $\mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}$.

Conversion to the equality constraints

To convert inequalities into equalities in the constraints, we introduce **slack** variables and **surplus** variables:

Slack Variables: To convert a constraint of “less than or equal to” type into equality type, we add a variable in the left hand side is known as slack variable.

Example: $4x_1 + 8x_2 - 3x_3 \leq 15 \implies 4x_1 + 8x_2 - 3x_3 + v_1 = 15$

Surplus Variables: To convert a constraint of “greater than or equal to” type into equality type, we subtract a variable in the left hand side is known as surplus variable.

Example: $x_1 + 9x_2 + 13x_3 \geq 21 \implies x_1 + 9x_2 + 13x_3 - v_2 = 15.$

Convert the following LPP into standard form:

Subject to:

$$\begin{aligned} \text{Max } Z &= x_1 - 4x_2 + x_3 \\ 2x_1 - 6x_2 + x_3 &\leq 6 \\ x_1 + 4x_2 &= 5 \\ 3x_1 - x_3 &\geq -10, \\ x_1 \geq 0, x_2 \geq 2, x_3 &\in \mathbb{R}. \end{aligned}$$

Solution: (Step I) Since $x_2 \geq 2 \Rightarrow x_2 - 2 \geq 0$. So, substitute $x_2 - 2 = x'_2$, where $x'_2 \geq 0$.
Again since $x_3 \in \mathbb{R} \Rightarrow x_3$ unrestricted in sign. So substitute $x_3 = x'_3 - x''_3$, where $x'_3, x''_3 \geq 0$.
With these substitutions, the give LPP reduced to

$$\begin{aligned} \text{Max } Z &= x_1 - 4(x'_2 + 2) + x'_3 - x''_3 \\ \text{Sub to: } 2x_1 - 6(x'_2 + 2) + x'_3 - x''_3 &\leq 6 \\ x_1 + 4(x'_2 + 2) &= 5 \\ 3x_1 - x'_3 + x''_3 &\geq -10, \\ x_1, x'_2, x'_3, x''_3 &\geq 0. \end{aligned}$$

(Step II) To make the right hand side of each constraint nonnegative, we multiply the second and third constraints by -1 .

$$\begin{aligned} \text{Max } Z &= x_1 - 4x'_2 + x'_3 - x''_3 - 8 \\ \text{Sub to: } 2x_1 - 6x'_2 + x'_3 - x''_3 &\leq 18 \\ -x_1 - 4x'_2 &= 3 \\ -3x_1 + x'_3 - x''_3 &\leq 10, \\ x_1, x'_2, x'_3, x''_3 &\geq 0 \end{aligned}$$

(Step III) Now convert the first and third constraint by introducing slack variables s_1 and s_2 , respectively.

$$\begin{aligned} \text{Max } Z &= x_1 - 4x'_2 + x'_3 - x''_3 + 0.s_1 + 0.s_2 - 8 \\ \text{Sub to: } 2x_1 - 6x'_2 + x'_3 - x''_3 + s_1 &= 18 \\ -x_1 - 4x'_2 &= 3 \\ -3x_1 + x'_3 - x''_3 + s_2 &= 10, \\ x_1, x'_2, x'_3, x''_3 &\geq 0 \end{aligned}$$

Convert the following LPP into standard form:

$$\text{Max } Z = 2x_1 + 3x_2 - 7x_3$$

Subject to:

$$x_1 + 2x_2 - x_3 \leq -16,$$

$$2x_1 + x_2 + 4x_3 \geq 10,$$

$$x_1 \geq 0, x_2 \leq 0, x_3 \text{ is unrestricted.}$$

Solution: (Step I) Since $x_2 \leq 0 \Rightarrow -x_2 \geq 0$. So, substitute $-x_2 = x'_2$, where $x'_2 \geq 0$.

Again since x_3 unrestricted in sign. So substitute $x_3 = x'_3 - x''_3$, where $x'_3, x''_3 \geq 0$.

With these substitutions, the give LPP reduced to:

$$\text{Max } Z = 2x_1 - 3x'_2 + 7(x'_3 - x''_3)$$

$$\text{Sub to: } x_1 - 2x'_2 - (x'_3 - x''_3) \leq -16,$$

$$2x_1 - x'_2 + 4(x'_3 - x''_3) \geq 10,$$

$$x_1, x'_2, x'_3, x''_3 \geq 0.$$

(Step II) To make the right hand side of each constraint nonnegative, we multiply the first constraint by -1 .

$$\text{Max } Z = 2x_1 - 3x'_2 + 7x'_3 - 7x''_3$$

$$\text{Sub to: } -x_1 + 2x'_2 + x'_3 - x''_3 \geq 16,$$

$$2x_1 - x'_2 + 4x'_3 - 4x''_3 \geq 10,$$

$$x_1, x'_2, x'_3, x''_3 \geq 0$$

(Step III) Now convert the first and third constraint by introducing slack variables s_1 and s_2 , respectively.

$$\text{Max } Z = 2x_1 - 3x'_2 + 7x'_3 - 7x''_3 + 0.s_1 + 0.s_2$$

$$\text{Sub to: } -x_1 + 2x'_2 + x'_3 - x''_3 - s_1 = 16,$$

$$2x_1 - x'_2 + 4x'_3 - 4x''_3 - s_2 = 10,$$

$$x_1, x'_2, x'_3, x''_3 \geq 0$$

Ex: Find the standard form of the given LPP

$$\text{Min } Z = \max\{|2x_1 + 3x_2|, |x_1 - 2x_2|\}$$

Solution: Let, $y = \max\{|2x_1 + 3x_2|, |x_1 - 2x_2|\}$.

$$\text{Then, } |2x_1 + 3x_2| \leq y \Rightarrow 2x_1 + 3x_2 \leq y, \quad 2x_1 + 3x_2 \geq -y,$$

$$\text{And } |x_1 - 2x_2| \leq y \Rightarrow x_1 - 2x_2 \leq y, \quad x_1 - 2x_2 \geq -y.$$

$$\text{Min } Z = y$$

Subject to:

$$2x_1 + 3x_2 - y \leq 0$$

$$2x_1 + 3x_2 + y \geq 0$$

$$x_1 - 2x_2 - y \leq 0$$

$$x_1 - 2x_2 + y \geq 0, \quad x_1, x_2, x_3 \geq 0.$$

Now we introduce slack variables s_1 and s_2 in the first and third constraints and surplus variables s_3 and s_4 in the second and fourth constraints.

$$\text{Min } Z = y$$

Subject to:

$$2x_1 + 3x_2 - y + s_1 = 0$$

$$2x_1 + 3x_2 + y - s_3 = 0$$

$$x_1 - 2x_2 - y + s_2 = 0$$

$$x_1 - 2x_2 + y - s_4 = 0,$$

$$x_1, x_2, x_3, s_1, s_2, s_3, s_4 \geq 0.$$

Basic Solutions

- Consider a system of m equations with n variables ($n \geq m$):
 $Ax = b$, where $x \in \mathbb{R}^n$.

A **basic solution** is obtained by:

- Select m variables (called **basic variables**).
- Setting the remaining $(n - m)$ variables (called **non-basic variables**) equal to zero.
- Solve the resulting system of m equations in m unknowns.
- Each such solution is called a basic solution.

Degenerate & Non-degenerate Basic Solutions

- A basic solution is called a **non-degenerate basic solution** if all the basic variables are strictly positive.
- A basic solution is called a **degenerate basic solution** if one or more of the basic variables take the value zero.

Example: Find the basic solutions of the following system:

$$x_1 + x_2 = 4, \quad x_2 + x_3 = 4.$$

Solutions: Here number of variables (n) = 3, number of equations (m) = 2.

So, the number of non basic variables $n - m = 3 - 2 = 1$.

$x_1 = 0$	$x_2 = 4, x_3 = 0$	$(0, 4, 0)$	Degenerate BFS
$x_2 = 0$	$x_1 = 4, x_3 = 4$	$(4, 0, 4)$	Non-degenerate BFS
$x_3 = 0$	$x_1 = 0, x_2 = 4$	$(0, 4, 0)$	Degenerate BFS

Ex: Find the basic solutions and categorized them as feasible or infeasible:

$$2x_1 + x_2 \leq 4; \quad x_1 + 2x_2 \leq 5; \quad x_1, x_2 \geq 0.$$

Solution: The standard form of the given system is:

$$2x_1 + x_2 + s_1 = 4,$$

$$x_1 + 2x_2 + s_2 = 5,$$

$$x_1, x_2, s_1, s_2 \geq 0.$$

Here number of variables (n) = 4, number of equations (m) = 2.

So, the number of non basic variables $n - m = 4 - 2 = 2$.

Non Basic Variables	Basic Variables	Basic solution	Feasible/Infeasible
$x_1 = 0, x_2 = 0$	$s_1 = 4, s_2 = 5$	$(0, 0, 4, 5)$	Feasible
$x_1 = 0, s_1 = 0$	$x_2 = 4, s_2 = -3$	$(0, 4, 0, -3)$	Infeasible
$x_1 = 0, s_2 = 0$	$x_2 = 5/2, s_1 = 3/2$	$(0, 2.5, 1.5, 0)$	Feasible
$x_2 = 0, s_1 = 0$	$x_1 = 2, s_2 = 3$	$(2, 0, 0, 3)$	Feasible
$x_2 = 0, s_2 = 0$	$x_1 = 5, s_1 = -1$	$(5, 0, -1, 0)$	Infeasible
$s_1 = 0, s_2 = 0$	$x_1 = 1, x_2 = 2$	$(1, 2, 0, 0)$	Feasible

One-to-one correspondence in the planar (graphical) case

For a linear program in two variables, the set of basic feasible solutions coincides exactly with the set of vertices of the feasible polygon obtained in the graphical method.

$BFS \Leftrightarrow \text{Vertex of the feasible region}$

Consequently, since the objective function is linear, if an optimal solution exists, then it is attained at one (or possibly more) of these BFS (vertices). However, not every BFS is optimal.

EX: Consider the following LPP and show the one-one correspondence between the BFS and the vertices of the feasible region.

$$\text{Max } Z = 2x_1 + 3x_2$$

Subject to: $2x_1 + x_2 \leq 4$; $x_1 + 2x_2 \leq 5$; $x_1 \geq 0$; $x_2 \geq 0$.

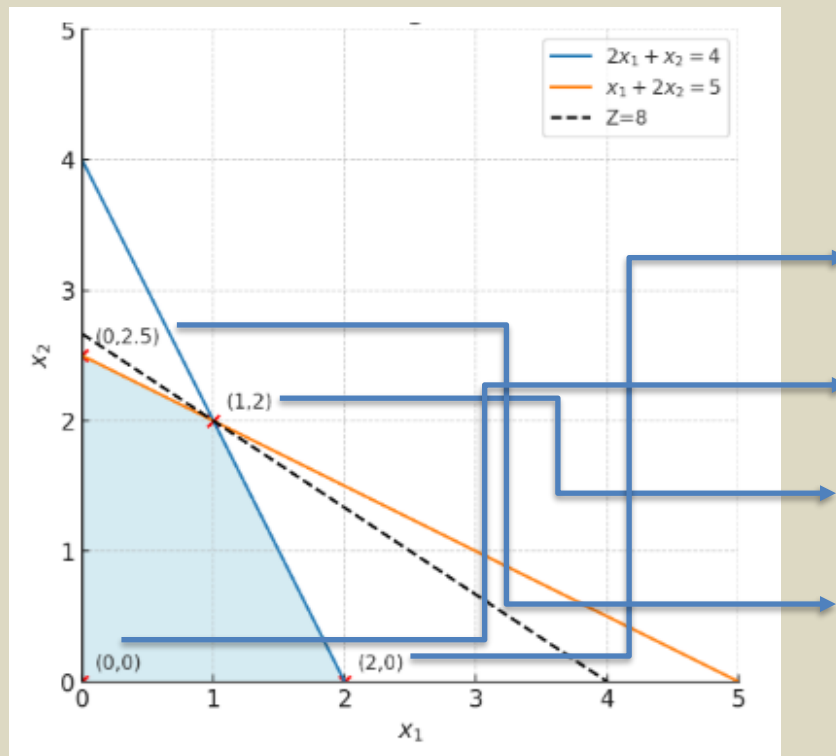
Solution: We introduce slack variables s_1 and s_2 in the first and second constraints.

$$\text{Max } Z = 2x_1 + 3x_2$$

Subject to: $2x_1 + x_2 + s_1 = 4$; $x_1 + 2x_2 + s_2 = 5$; $x_1 \geq 0$; $x_2 \geq 0$, $s_1 \geq 0$, $s_2 \geq 0$.

Proceed similar as previous example, the set of all BFS is

$$\{(0, 0, 4, 5), (0, 5/2, 3/2, 0), (2, 0, 0, 3), (1, 2, 0, 0)\}.$$



Vertices	Basic Feasible solution	Value of Z
(0, 0)	(0, 0, 4, 5)	0
(2, 0)	(2, 0, 0, 3)	4
(1, 2)	(1, 2, 0, 0)	8
(0, 5/2)	(2, 0, 0, 3)	15/2

Non-basic solutions:

A **Non-Basic Solution** is just any feasible solution that is **not a basic solution**.

Example: For the system: $x_1 + x_2 = 4, \quad x_1, x_2 \geq 0$

Basic Solutions:

- Take x_1 as basic variable, $x_2 = 0, \rightarrow (4, 0)$ is a BFS.
- Take x_2 as basic variable, $x_1 = 0, \rightarrow (0, 4)$ is a BFS.

Non-basic solutions: Any other feasible point, e.g., $(2, 2)$, which satisfies the equation but is not derived by setting one variable to 0.

Ex: Find a non basic solution of the system using **row reduction**:

$$x_1 - 2x_2 + x_3 = 2, \quad x_1 + x_2 - x_3 = 4.$$

Solution: The given system can be represented in the augmented form:

$$\begin{bmatrix} 1 & -2 & 1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

The **augmented matrix** is given by:

$$[A: b] = \begin{bmatrix} 1 & -2 & 1 & : & 2 \\ 1 & 1 & -1 & : & 4 \end{bmatrix} \xrightarrow{R'_2 = R_2 - R_1} \begin{bmatrix} 1 & -2 & 1 & : & 2 \\ 0 & 3 & -2 & : & 2 \end{bmatrix} \xrightarrow{R'_1 = R_1 - \frac{2}{3}R_2} \begin{bmatrix} 1 & 0 & -\frac{1}{3} & : & \frac{10}{3} \\ 0 & 3 & -2 & : & 2 \end{bmatrix}$$

$$\xrightarrow{R'_2 = \frac{1}{3}R_2} \begin{bmatrix} 1 & 0 & -\frac{1}{3} & : & \frac{10}{3} \\ 0 & 1 & -\frac{2}{3} & : & \frac{2}{3} \end{bmatrix}$$

The reduced system: $x_1 - \frac{x_3}{3} = \frac{10}{3}, x_2 - \frac{2x_3}{3} = \frac{2}{3}$

Put $x_3 = 2, \Rightarrow x_1 = 4, x_2 = 2.$

Therefore, **$(4, 2, 2)$ is a non basic solution.**

Exercises

Ex 1: After converting the standard form, find all possible basic solutions of the following LPP:

$$\begin{aligned} \text{Max } Z &= x_1 + 3x_2 \\ \text{Subject to: } \quad &x_1 + x_2 \leq 2, \\ &-x_1 + x_2 \leq 4, \quad x_1, x_2 \geq 0. \end{aligned}$$

Ex 2: Find all possible basic solutions and categorized them in degenerate or no degenerate BFS of the following LPP:

$$\begin{aligned} \text{Max } Z &= x_1 + 2x_2 - 3x_3 - 2x_4 \\ \text{Subject to: } \quad &x_1 + 2x_2 - 3x_3 + x_4 = 4, \\ &x_1 + 2x_2 + x_3 + 2x_4 = 4, \quad x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

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